

Lecture Notes

Course: Complex Analysis

TA: Morales C. Ronny

Topic: Convergence Tests

Example 4.1 (Geometric Series). Consider the series generated by the sequence $(z^n)_{n=0}^\infty$, then

$$|s_k - L| = \left| \sum_{k=0}^n z^k - \frac{1}{1-z} \right| = \frac{|z|^{n+1}}{|1-z|},$$

if we take

$$N(\varepsilon, z) = \frac{\ln(\varepsilon|1-z|)}{\ln|z|} - 1 \quad \left[\text{respectively, } N(\varepsilon) = \frac{\ln \varepsilon(1-r)}{\ln r} - 1 \right],$$

then for any $n > N$, $|s_k - L| < \varepsilon$ so that the series converges pointwise [respectively, uniformly]. $\square$

Example 4.2. c) Use partial fraction's decomposition to get

$$\sum_{n=1}^m \frac{1}{n(n+1)} = \sum_{n=1}^m \left(\frac{1}{n} - \frac{1}{n+1} \right) = s_m = 1 - \frac{1}{m+1},$$

so that $\lim s_m = 1$. $\square$

Proof 4.4 (The n th term test). If the series converges to, say, S , then the partial sums are such that

$$\lim s_n = S \implies z_n = \lim s_n - \lim s_{n-1} = S - S = 0.$$

Proof 4.5 (Leibniz Criterion for Series). ($\implies$) Corollary of the n th term test.

($\Leftarrow$) Suppose $\lim x_n = 0$ with (x_n) a nonnegative decreasing sequence, then for the even partial sums we have

$$S_{2m} = x_1 - x_2 + x_3 - \cdots - x_{2m} = \sum_{k=1}^m (x_{2k-1} - x_{2k}),$$

which is a sum of nonnegative terms since $x_{2k-1} - x_{2k} \geq 0$ (the sequence is decreasing) and then we can rearrange differently to notice how

$$S_{2m} = x_1 - \sum_{k=1}^{m-1} (x_{2k} - x_{2k+1}) - x_{2m} \leq x_1,$$

again, because $x_{2k} - x_{2k+1} \geq 0$. We conclude that S_{2m} is bounded above and increasing (sum of nonnegative terms) so that it is convergent. Moreover, for the odd partial sums we have

$$S_{2m-1} + (-x_{2m}) = S_{2m} \iff S_{2m-1} = S_{2m} + x_{2m} \implies \lim S_{2m} = \lim x_n + \lim S_{2m} = \lim S_{2m}$$

where in the last equality we used $\lim x_n = 0$. We conclude that $\lim S_n = 0$ and thus the series converges. $\square$

Proof 4.11 (Comparison Test). Suppose that there is some $K \in \mathbb{N}$ such that

$$0 \leq x_n \leq y_n \quad \text{for } n \geq K,$$

then if $\sum y_n$ is convergent we have that for any $\varepsilon > 0$ there is some natural G such that

$$|s_n^y - s_m^y| < \varepsilon \quad \text{for } n, m \geq G,$$

now choose $M := \max(K, G)$ so that

$$|s_n^x - s_m^x| = \left| \sum_{k=n+1}^m x_k \right| = \sum_{k=n+1}^m x_k \leq \sum_{k=n+1}^m y_k = |s_n^y - s_m^y| < \varepsilon \quad \text{for } n, m \geq M,$$

and hence $\sum x_n$ is convergent by the Cauchy criterion.

This shows i) and ii) is the contrapositive of i). $\square$

Proof 4.18. ($\Rightarrow$) Suppose $X = (x_n)$ is convergent to, say x , then for any $\varepsilon > 0$ there is some $K \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon/2 \quad \text{for } n \geq K,$$

which is equivalent to

$$x - \varepsilon/2 < x_n < x + \varepsilon/2,$$

so $x + \varepsilon/2$ is an upper bound [respectively, $x - \varepsilon/2$ is a lower bound] for the K -tail X_K of X and thus

$$\sup X_K \equiv u_K \leq x + \varepsilon/2 \quad [\text{respectively, } \inf X_K \equiv \ell_K \geq x - \varepsilon/2]$$

so that by theorem 2.6 we get

$$\lim u_K \equiv \limsup x_n \leq x + \varepsilon/2 \quad [\text{respectively, } \lim \ell_K \equiv \liminf x_n \geq x - \varepsilon/2],$$

and finally, using $\liminf x_n \leq \limsup x_n$ we get

$$x - \varepsilon/2 \leq \liminf x_n \leq \limsup x_n \leq x + \varepsilon/2,$$

but since this holds for arbitrary ε , we conclude that

$$\liminf x_n = \limsup x_n = x.$$

($\Leftarrow$) Conversely, suppose that $\liminf x_n = \limsup x_n = x$, then for any $\varepsilon > 0$ there are naturals K_1, K_2 such that

$$|u_m - x| < \varepsilon \quad \text{and} \quad |\ell_m - x| < \varepsilon, \quad u_m := \sup X_m, \quad \ell_m := \inf X_m, \quad (\star)$$

for $m \geq M := \max(K_1, K_2)$, and we can now see that by definition of u_m and ℓ_m we have

$$\ell_m \leq x_n \leq u_m \quad \text{for } n > m, \quad (\star\star)$$

so that $(\star)$ and $(\star\star)$ together imply that

$$x - \varepsilon < \ell_m \leq x_n \leq u_m < x + \varepsilon \implies x - \varepsilon < x_n < x + \varepsilon \iff |x_n - x| < \varepsilon \quad \text{for } n > m.$$

We therefore conclude that $\lim x_n = x$. $\square$

Lemma.